

Information provision and externalities: The impact of genomic testing on the dairy industry

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Abstract

We estimate the impact of a new information technology, genomic selection, on the market for dairy cattle genetics in the United States. Genomic selection allows cattle breeders to test new cattle as soon as they are born for the presence of desirable, productive traits and more than double their pace of genetic improvement in milk production. At the same time, the rate of inbreeding, the extent to which dairy bulls are related to one another, greatly increased after the introduction of genomic testing. Using a data set of bulls marketed in the US from 2000 to 2020, we examine whether inbreeding rates increased due to increased competition for popular dairy cattle families in the face of greater market competition. We use a difference-in-difference and matching approach and find that inbreeding increased greatly in already popular families after the introduction of genomic testing relative to less popular families with similar production and health traits. Our estimation shows that the increased inbreeding rate of American bulls caused the entire industry to lose between 3.6 and 6.7 billion dollars from 2011 to 2019.

1 Introduction

The US plays a crucial role in the global dairy sector as one of the the world's largest suppliers of dairy bull genetics. In 2022, the United States alone exported \$295 million of bovine semen, representing 47% of the world's exports in value terms (Figure 1). For about six decades, milk production per cow has consistently increased in the US herd, and it is estimated that more than half of this growth can be attributed to improvements in cattle genetics (Council on Dairy Cattle Breeding, 2025). Thus, the efforts of US breeders have likely had substantial impacts on the global dairy industr.

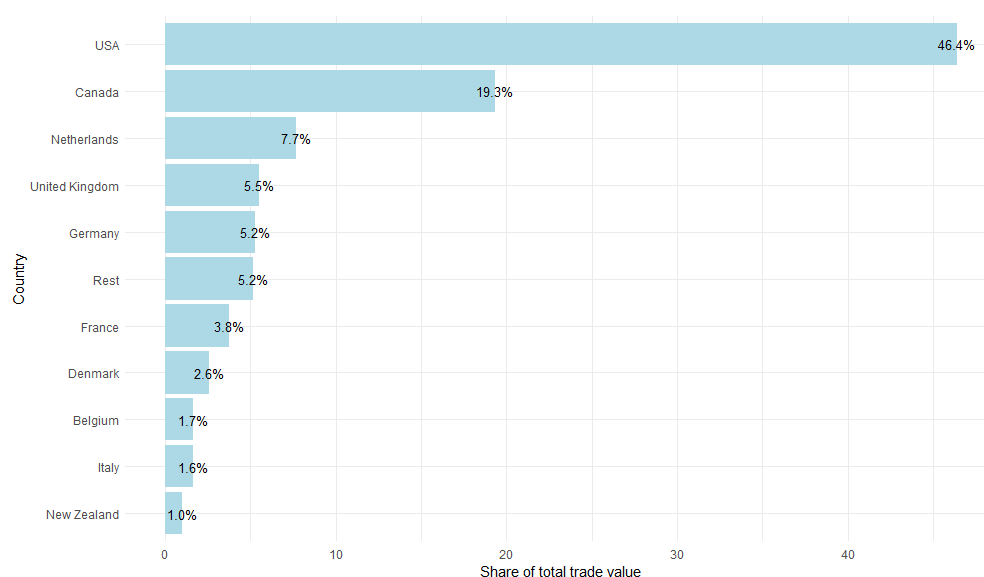


Figure 1: Share of trade Value of Bovine Semen exports by country (2019)

Source: UN Comtrade database

US dairy breeders improve cow milk yield by identifying which bulls have the most productive female offspring and then continuing to breed off these lines. The speed of this genetic improvement has previously been limited by biological factors. Determining whether a bull had good milk production ability took **at least three years** since the bull had to have offspring born from their use on dairy farms, and those offspring would need to take two years to grow to production age.

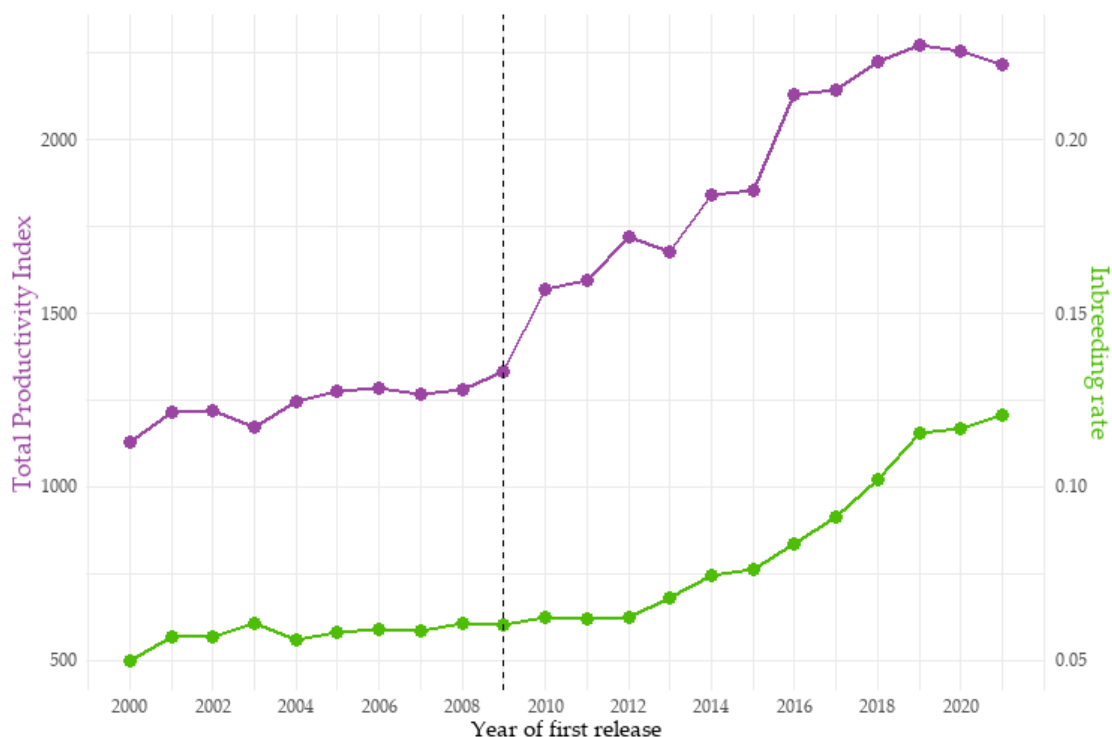
However, these limits became less binding after the advent of an innovation in testing

technology: genomic testing. Introduced in 2008, genomic testing allows breeders to link each animal's genes to productive outcomes such as milk yield and fat yield. The practical upshot is that determining a dairy bull's quality can now be done as soon as it is born instead of in **three** years. There are two important, documented impacts of genomic testing on the dairy bull market. First, the pace of genetic improvement has more than doubled in dairy cattle since breeders can find productive bulls at much younger ages and breed new generations even quicker. Second, dairy companies have more than **doubled the number of bulls** for sale on the market.

However, the introduction of genomic testing has coincided with a less desirable outcome: an increase in inbreeding, the degree to which a dairy bull is related to the rest of the dairy cattle population. Figure 2 shows the average inbreeding rate of dairy bulls on the market alongside the **total productivity index** (TPI), a measure of genetic **milk productive** ability. Inbreeding is an example of a negative externality in the cattle genetics industry since higher inbreeding rates expose the industry to more disease risk and depressed yields in future generations, costs not borne directly by the breeder in the short run. Since dairy breeders continually breed from each other's genetic stock, the breeding decisions of one breeder can affect all other breeders through the shared public resource that is the US cattle genetic pool. In one case, breeding excessively from a famous dairy bull "Pawnee Farm Arlinda Chief" led to more than 500,000 spontaneous abortions and nearly 420 million dollars in losses because of an undesirable, recessive trait that was passed to his offspring (Adams et al., 2016). Given the US's role in the global cattle genetics market, **damaging** the diversity of the dairy cattle gene pool could also damage livestock markets worldwide.

In this paper, we examine whether the introduction of genomic testing increased inbreeding by increasing competition for already popular dairy bull genetic lines. Since genomic testing expanded the number of bulls on the market and quickened the pace of genetic improvement, we test the theory that these competitive forces led animal breeders

Figure 2: Dairy Bull Productivity versus Inbreeding Rate



to increase market share by crossing already popular lines instead of finding new sources of genetics. To isolate the role of popularity, we construct a set of counterfactual bulls that are comparable to popular genetic families in terms of productivity using propensity score matching. Using a difference-in-difference approach, we then examine inbreeding rates in popular families relative to our counterfactual families before and after the introduction of genomic testing in 2010. If genomic testing increases competition for quality lines independent of popularity among dairy farmers, we expect that inbreeding rates for these two groups will be the same before and after its introduction.

We find that popular dairy bull lines became more inbred after the introduction of genomic testing than our counterfactual. The inbreeding rate of the most popular lines increased by 1-2.8 percentage points relative to our control group after 2010. Before genomic testing, there was no statistically significant difference in inbreeding rates across our groups. Using a rough estimate of the costs of inbreeding from the breeding association Holstein USA, a back-of-the-envelope calculation implies that this increase in inbreeding

led to approximately \$3.6-6.7 billion in losses to the industry from 2011 to 2019. Since dairy breeders are incentivized to inbreed for short-run gains even at the expense of their genetic pool, this indicates a role for regulation to penalize the industry for inbreeding in the short run to prevent damages to the industry in the long run (Cole, 2023).

This article fills a critical gap in the literature on livestock markets and the behavior of animal breeders. Few economics papers analyze the strategic actions of animal breeders and even fewer analyze how their actions may impose costs on the industry. Early modeling of the animal breeder's decision dates back to studies such as Ladd and Gibson (1978), Kerr (1984), and Melton et al. (1994) which use hedonic regression and linear programming to determine the implicit market value of different livestock traits. Richards and Jeffrey (1996) and Schroeder et al. (1992) apply this method to dairy bulls to determine the economic value of each dairy bull trait implied by the results of a hedonic regression. Thus far, the economics literature has not appreciated how the decisions of animal breeders may lead to worse outcomes for the whole sector.

However, a related literature in agricultural economics has explored similar coordination problems among farmers controlling pests or disease. In the short-run, farmers are incentivized to forgo the costs of controlling a future pest or disease because of a strong incentive to **free-ride** off of the efforts of others (Epanchin-Niell and Wilen, 2015). In the planting of Bt-resistant crops, farmers may be incentivized to plant Bt-resistant corn or cotton and inadvertently cause the emergence of Bt-resistant insects in the future (Brown, 2018, Livingston et al., 2004, Wechsler and Smith, 2018). Thus, farmers impose a negative externality on their surrounding area when they do not plant non-Bt crops on some proportion of their land to reduce resistance in the future (Brown, 2018, Desquilbet and Herrmann, 2016, Huang et al., 2011, Hurley et al., 2001). Similarly, Lence and Singerman (2023) finds that there are significant coordination problems to overcome in dealing with a disease like citrus greening.

Our paper is the first to explore similar coordination problems in the market for dairy

cattle genetics. Breeders are incentivized in the short run to increase inbreeding by breeding cattle from famous lines, which may boost genetic improvement and capture more market share. However, inbreeding causes a negative externality by increasing the chance of harmful recessive traits and weakening the genetic pool's resiliency. Since animal breeders do not directly pay the cost of inbreeding in the short run, the expected result is a level of inbreeding that is higher than should be ideal for the industry. Animal scientists have theorized that this increase in inbreeding is due to breeders using already popular genetics lines to gain a competitive edge (Cole, 2019).

Before its introduction, it was theorized that genomic testing would **decrease** inbreeding by allowing breeders to find productive dairy bulls outside known genetic lines more easily (Daetwyler et al., 2007). Instead, we find that the average inbreeding coefficient of dairy bulls increased along with dairy bull productivity. We provide evidence of this theory by comparing the inbreeding rates of groups of bulls that are similar in genetic traits but different in popularity before and after genomic testing. If genomic testing led to an increase in the discovery of less popular lines, we should expect no difference in the inbreeding rates of these groups since animal breeders would have selected them only on the basis of their genetic traits. Instead, we find that genetic lines already popular before genomic testing experienced an increase in the average inbreeding rate, leading to significant economic damage to the dairy sector.

While we estimate that inbreeding led to as much as 6.7 billion dollars in damages for the sector, **we also find that it is likely that the increases in milk productivity made up for the damages**. We estimate that the net benefits of genomic testing may have been as high as 10.4 billion dollars due to an increase in **milk production**. However, a decrease in the resiliency of the US genetic pool may have hidden costs and global implications. Should a disease spread through inbreeding, it is difficult to account for how destructive and costly it might be. Worse, US dairy breeders will have impacted dairy herds around the world since US genetic lines have become so important to the global dairy industry.

The paper is structured as follows: Section 2 presents background and a conceptual model exploring how introducing genomic selection can lead to higher inbreeding rates in subsequent generations of dairy bulls. Section 3 explains the data sources and shows descriptive statistics. Section 4 outlines the identification strategy and methodological framework. In section 5, we explain the results before doing a cost-benefit analysis in section 6. We conclude in section 7 and discuss the implications for the future of the US dairy industry.

2 Conceptual Framework

2.1 The Animal Breeders' Problem

To produce the next generation of milking cows, dairy farms must choose a bull to breed with their current cows. Since breeding in the dairy sector is almost entirely done through artificial insemination, dairy farmers buy semen from a bull that is likely to produce offspring with good productivity and health. Dairy farmers determine this by looking at the genetic traits of each bull, which are the predicted performance of their female offspring. These genetic traits are calculated by the Council on Dairy Cattle Breeding (CDCB) using data on dairy cows enrolled in Dairy Herd Improvement Associations (DHIAs) (Hutchins and Hueth, 2023). Once a bull has enough offspring in the DHIA system, the CDCB will release this trait publicly to the industry. Because of the biological lags of cattle production, it takes roughly five years for a dairy bull, once born, to have enough data for a “daughter-proven” evaluation.

The supply side of the dairy bull semen market is made up of animal breeders who supply a product which is, in essence, an **inseparable bundle of genetic traits** Melton et al. (1979) and Melton et al. (1994) model the breeder’s decision as being paid based on how much higher their bull’s traits are than the population average, **which views all bulls as interchangeable to farmers apart from their genetic traits**. To some degree, this makes

animal breeding a zero-sum game. To maximize profits, breeders must release bulls into the market with the highest possible trait values to grab market share from other breeders. To market semen to farmers, breeders sell their semen through dairy genetics companies which are either cooperatives owned by dairy farmers or investor-owned. On the demand side, dairy farmers choose bulls from the market to maximize their own profits. Since bull semen is shipped frozen, dairy farmers have a significant scope of choice of bulls which is not restricted by where the bull lives.

When mating animals to produce new bulls to sell, breeders can decide how related they want the bull and cow to be. If the bull and cow are closely related, this is called “line breeding” since the breeder is making crosses within the same line or genetic lineage. Crossing a bull and a cow that are more distantly related is often referred to as “cross breeding.” The advantages of line breeding are that it allows breeders to increase the probability that the traits of their common descendant will continue to be expressed in future generations. For example, if a bull and a cow are both closely related to a bull with good traits, then the good traits are more likely to be expressed. As a result of line breeding in dairy farming, nearly all Holstein dairy bulls can be traced back to three dairy bulls born in the 1960s (Yue et al., 2015).

The disadvantage of line breeding is that it also increases the probability that the descendants will express a recessive trait that causes damage. Since the goal of linebreeding is to keep the degree of relationship with respect to a relative constant across the generations of his descendants, it leads to higher inbreeding rates. As the inbreeding rate of the dairy cow population increases, the likelihood of unforeseen genetic traits that could harm the industry also rises. In the short run, trait gain in an inbred population will also be lower because of “inbreeding depression” (Gutiérrez-Reinoso et al., 2022, Smith et al., 1998).

The primary advantages of cross-breeding is lowering the probability of undesirable recessive traits in the long run. Its primary disadvantage, however, is a potentially slower

gain in genetic traits. If the existing lines have desirable production ability, cross-breeding can dilute production ability by introducing new genetic material instead of honing in on one line's benefits.

When deciding to line breed or cross breed, breeders have to decide whether to benefit in the short run with potential long-run costs to their genetic lines versus foregoing short-run gains in genetics to increase the resiliency of their lines. The long-run costs of line breeding, however, are not directly borne by the breeder. While breeders will experience devaluation in the genetic traits through inbreeding depression, they will not necessarily pay the cost of increasing the probability of damaging recessive traits in their lines. Since breeders also use each other's lines, propagating recessive traits is a cost imposed on the entire industry.

From the perspective of an efficient social planner, there is a certain efficient level of cross-breeding in dairy bulls to continue genetic gain while maintaining genetic diversity at a safe level. A similar solution was developed in the planting of Bt crops to control resistance to the European corn bore. Starting in 1995, the Environmental Protection Agency in the US government began mandating that farmers plant "refuge" seed without Bt resistance alongside Bt resistant seed (Bourguet et al., 2005). The purpose of mandating refuges is to allow insects that are not resistant to Bt to survive and breed with Bt-resistant insects and reduce the level of resistance in the insect population. While farmers are individually incentivized to control Bt resistance, they do not completely internalize the costs of planting too little refuge seed since resistant insects can spread to other farms. Private seed companies now help farmers comply with this regulation by mixing non-resistant seed into the bags of seed they sell (called "refuge-in-a-bag") (Mohan Komarlingam, 2018).

Similar to the planting of Bt corn, dairy breeders are not necessarily incentivized to maintain a level of genetic diversity in their industry. In fact, the zero-sum nature of animal breeding described by Melton et al. (1994) would incentivize dairy breeders to invest too little in genetic diversity if the marginal returns to line breeding are higher in

the short run. This is because if a breeder decides to cross-breed when the rest of the industry line breeds, they will lose future market share by delivering less genetic gain than the rest of the industry. Moreover, if some dairy farmers prefer specific lines over others, breeders can capture market share by continuing to breed off of the same lines to maintain the purity of the genetic line and the farmer's "brand loyalty."

2.2 The Impact of Genomic Testing

The impact of genomic testing on the decisions of dairy breeders is best seen with a stylized version of the "breeder's equation" for a given genetic trait (Kerr, 1984):

$$\text{Trait Improvement} = \frac{\text{Heritability} \times \text{Selection Intensity}}{\text{Generation Interval}} \quad (1)$$

where:

- **Trait Improvement** is the change in that trait (e.g. milk production) in the new generation compared to the previous generation.
- **Heritability** measures the degree to which an offspring can "inherit" the trait from their parents.
- **Selection Intensity** measures the difference between the traits of the animals selected for breeding and those of the general population, representing how "selective" breeders are in the new generation.
- **Generation Interval** is the amount of time between generations, which is determined by how long it takes to determine the traits of the new generation.

The breeder's equation demonstrates that the rate at which genetic traits change from animal breeding is determined exogenously by the heritability of the trait but endogenously by i) how good breeders are at finding bulls with better traits (selection intensity)



and ii) how fast they can “prove” that bull to use for breeding more future bulls (generation interval).

Genomic testing increases trait improvement by decreasing the generation interval and increasing selection intensity. By using DNA samples, the potential impact of the bull on its offspring can be estimated by taking a DNA sample when it is born. With an accurate enough assessment, the bull can then be marketed and used as soon as it reaches sexual maturity, **around two years old**. This is a significant reduction in the generational interval, as bulls would not be marketed until they were around five years old using offspring data. Also, by having a better idea of what a bull’s traits are when its born, breeders can select more precisely the bulls that have the traits they want. This causes an increase in selection intensity which increases the rate of trait improvement.

The impact of genomic testing on inbreeding depends on how it changes the relative gains from line breeding or cross breeding for animal breeders. If genomic testing increased the relative gain of both options equally, there should be no additional increase in inbreeding from genomic testing. However, there are two ways that genomic testing could increase the relative gain of line breeding. First, genomic testing could have made selection more accurate for popular lines than it did less well-known lines. This would mean the selection intensity of line breeding would have increased more than it would have for cross-breeding, making the gain in traits higher in line breeding. Second, the increased number of bulls from genomic testing may have made popular lines more valuable as a way to distinguish bulls in a crowded market. While in Melton et al. (1994) bulls are interchangeable if they have the same traits, “brand loyalty” may manifest in the dairy bull market if there is an affinity for certain families of bulls. By line breeding these “super sires,” the breeders could capture market share from their popularity alone.

3 Data Description

The National Dairy Herd Improvement Program (NDHIP) collects, manages, and analyzes data from American dairy cattle (Hutchins and Hueth, 2023). The information it collects assesses the industry’s productivity and estimates the paternal contribution to these traits. The raw data collected from farms is used to estimate a statistical model to predict the genetic contribution of an animal as a sire (or dam) after controlling for environmental variables such as herd, age, pedigree, and season effects. The predicted values for each trait from this model are commonly called the **predicted transmitting ability** (PTA) of a trait.

The National Association of Animal Breeders (NAAB), a trade organization representing all of the major livestock genetics companies selling dairy bulls in the United States, publishes the genetic traits of all bulls being sold by their members. Traits are posted three times a year at the same time that each bull’s predicted genetic traits are calculated and posted publicly by the Council on Dairy Cattle Breeding (formerly by the USDA) (Hutchins and Hueth, 2023). Our data comes from the NAAB’s published lists from 2001 to 2020 and represents over 15,400 dairy bulls, both foreign and domestic, sold by NAAB members during this period. The NAAB represents a substantial portion of the livestock genetics companies and cooperatives in the US and Canada, making these data broadly representative of what is sold to US dairy farmers.

The genomic-testing program began in 2008, and the first genomically-tested bulls had evaluations published in January 2009. García-Ruiz et al. (2016) finds that, after 2010, the generation interval decreased significantly by about four years. Using the list of bulls that are actively sold by members of the National Association of Animal Breeders, we can see in Figure 3 that the new genomically proven bulls, first listed in 2010, have an average age of a little less than two years old when they are first released versus around five for daughter-proven bulls.

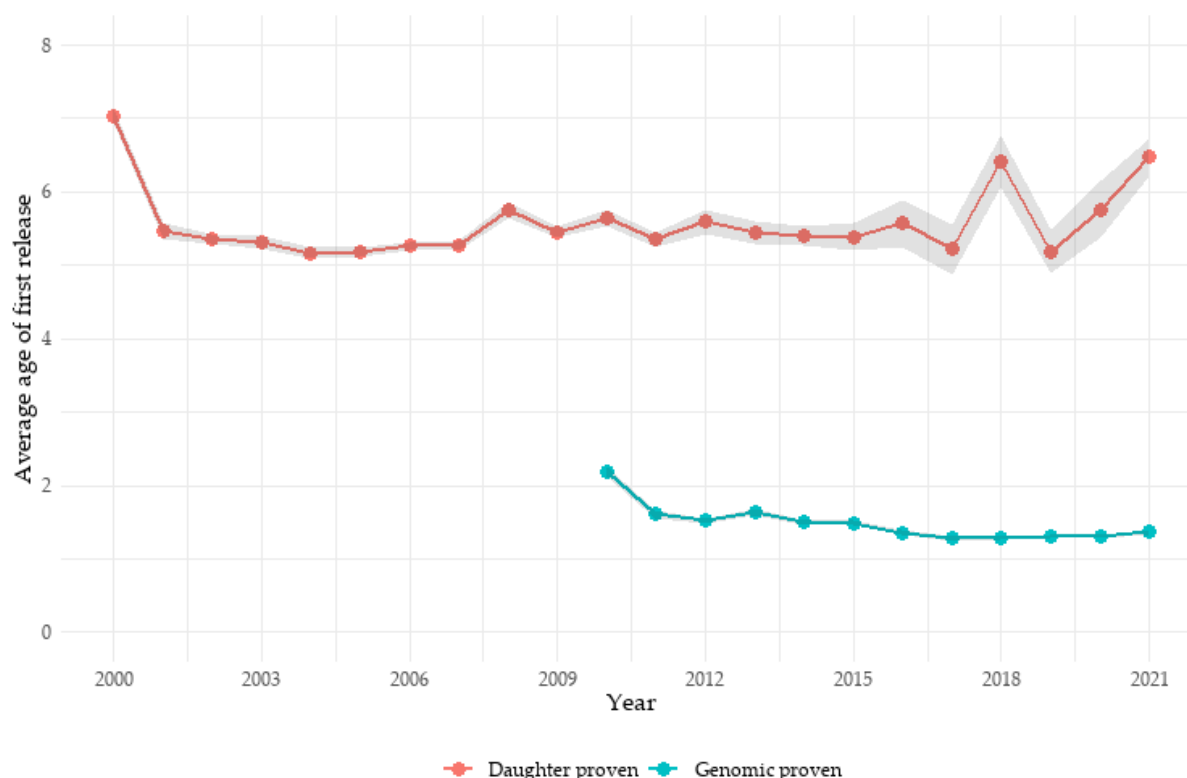


Figure 3: Average age at first proof for Holstein bulls
Source: NAAB

Another impact of genomic testing is an increase in the number of bulls being marketed. Figure 4 shows that the number of bulls listed as actively sold in the industry tripled starting in 2010. A large increase in the number of bulls implies that search costs to farmers may have increased when deciding which animal to buy genetics from (Farley, 1964, Stigler, 1961). In the face of higher search costs, brand loyalty to certain lines may be a way for dairy farmers to reduce their search costs. As seen in the plot, genomically-tested bulls soon became the dominant type of testing in the dairy genetics industry.

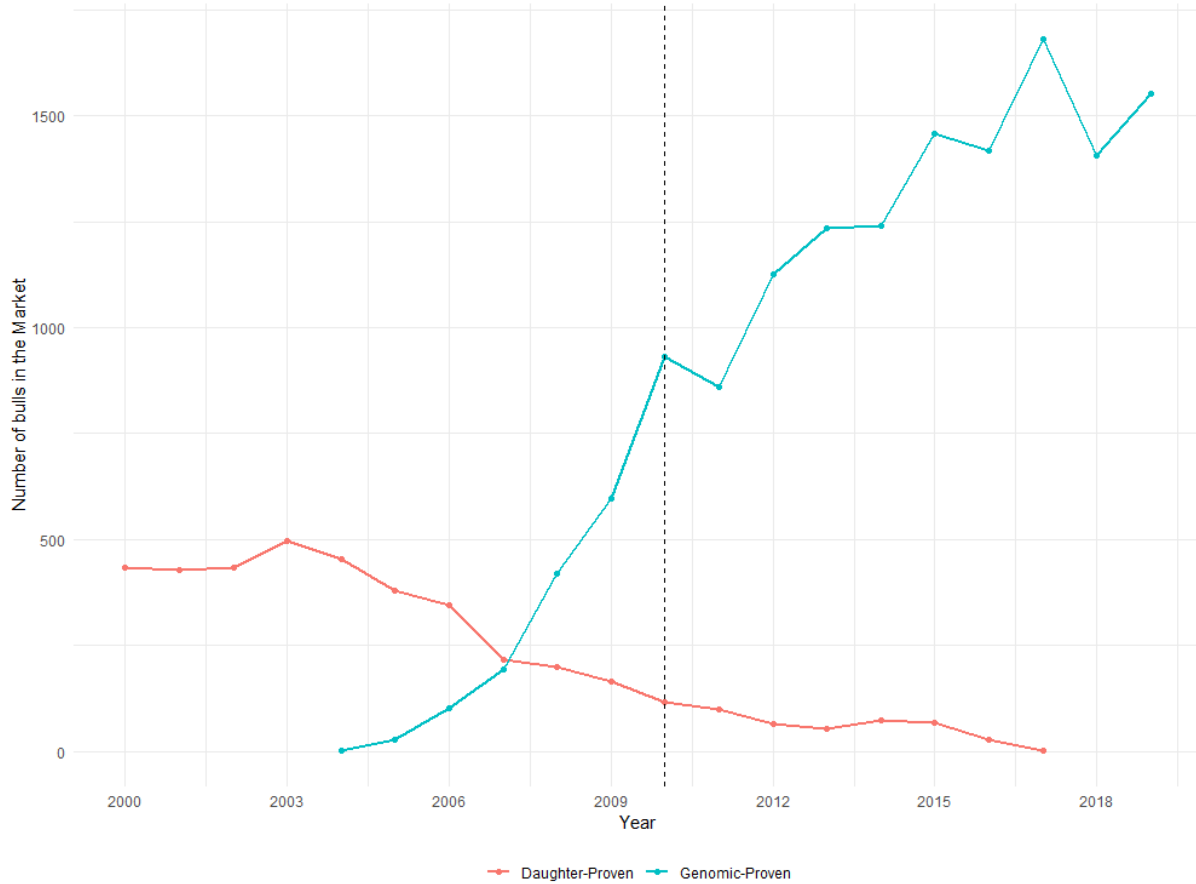


Figure 4: Total number of new bulls released in the market **Source:** NAAB

4 Empirical Framework

The goal of this analysis is to understand whether genomic testing increased inbreeding in cattle because of a preference for linebreeding already popular families of cattle. Since the percentage adoption of genomics is so high in the data, it is unlikely that we can use bulls that are not genomically tested as a control group. Moreover, bulls that are not genomically tested are more likely to have better quality traits and be more widely used. As an empirical approach, we must distinguish between breeders focusing efforts on these lines because of their existing quality versus their existing popularity, which would give higher genetic gain to breeders.

To isolate the role of the existing popularity of genetics, we construct a counterfactual

sample that has the same genetic traits but is from a less popular line. Specifically, we use propensity score matching (Smith and Todd, 2005) to define a control group that resembles the treatment in all observable characteristics, in this case the genetic traits. If genomic testing led to a race for quality bulls regardless of their previous use, we should see the inbreeding rate of these lines increase the same amount as the popular lines. If, instead, genomic testing accelerated line breeding for the popular lines, we should see a significant difference in the inbreeding rates after 2010.

We define a **line** as the set of direct (male) descendants from a particular bull. We also define a “super star” as a bull whose number of sons is in the 95th percentile of the distribution of offspring. To construct the line, we start by considering sires born between 1998 and 2004, as the “founders” of each line. Our definition depends critically on when the founders of such lines were born; if we choose too early of a date, lines are too difficult to distinguish since all bulls are related to some degree. After identifying the sires on which to build lines, we recursively build their pedigrees by matching bulls with their sires until we find the latest animal in the sample. Figure 14 in the appendix summarizes the sample selection mechanism; the treatment and control groups are comprised of the descendants of two groups: supersires (treatment) and matched sires (control).

The first step is to match the founders of such lines with other sires that did not father as many sons but had comparable genetic traits. Second, we find all descendants from both groups and compare the evolution of their inbreeding rates through time. The idea behind this procedure is that when those bulls were in the market, any reasonably similar sire could have been chosen but was not, and thus, can be used to construct a counterfactual line. To do so, we use a propensity score matching estimator, let $p(X) = \Pr(T_j = 1|X)$ where T is a dummy equal to 1 if sire j 's offspring is on the 95th percentile of the distribution and X_j be a vector comprised of the following:

1. Predicted Transmitting Ability for production traits (milk, fat, and protein), health traits (mastitis, metritis, productive life), and reproductive traits (daughter preg-

nancy rate, **effective calving**, gestation length).

2. Dummies for the absence of certain recessive genetic disorders (Holstein Foundation, 2021) such as *brachyspina*, cholesterol deficiency, and mule foot.
3. Dummies for the presence of genes associated with specific qualitative traits (polled, red coat color, **complex vertebral malformation**).
4. Dummies for the presence of haplotypes (specific sequences of DNA at different locations in the chromosome) that affect fertility (Holstein Foundation, 2018) when present in homozygous form (likely to be expressed when two carrier animals are mated).

We then estimate the propensity score (Cameron and Trivedi, 2005, p. 873), the probability of being a superstar given the level of PTAs and the presence of the entire set of variables listed above. The validity of our method relies on a series of assumptions (Heckman et al., 1997):

Assumption 1 (Stable Unit Treatment Value Assumption): $(inb^{treatment}, inb^{control}) \perp\!\!\!\perp T/X$.

The average inbreeding rates of any two lines do not vary with the treatment assigned to other lines. For each unit, there are no different forms of each treatment level (Imbens and Rubin, 2015), conditional on a set of covariates. This assumption implies that conditional on X , the distribution of inbreeding rates for both groups will be the same for all founders, but it must differ for their offspring. Figure 5, in the Appendix, shows the extent to which the inbreeding rates distribution of our matched sample resembles that of the treatment group. Unlike unmatched sires, their inbreeding distribution has a similar mean and variance conditional on the set of genetic traits.

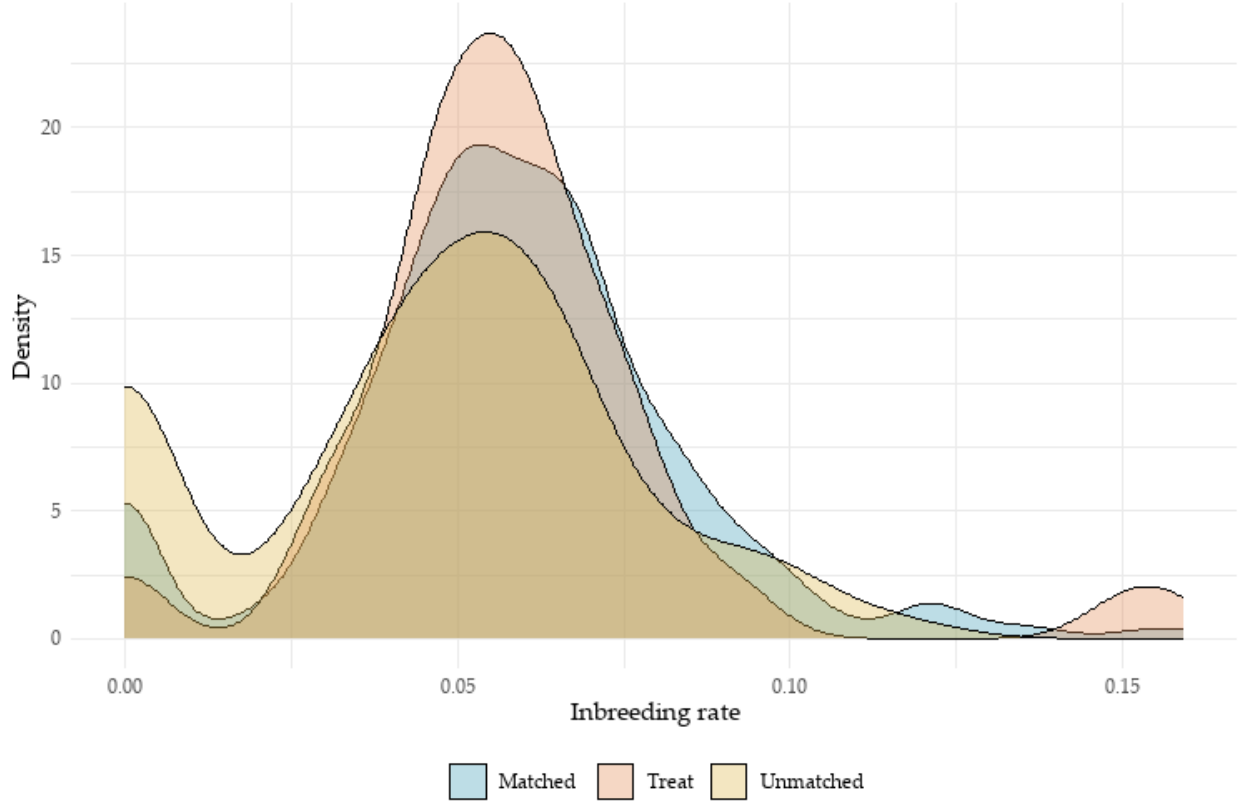


Figure 5: Distribution of Inbreeding rates across pre-treatment groups

Assumption 2: $0 < \Pr(T_j = 1 | \mathbf{X}) < 1$ for all $\mathbf{X}$

This assumption implies that we can define a propensity score for every value in the treatment group. Assumptions 1 and 2 together are the “strong ignorability” assumption in Rosenbaum and Rubin (1983), and they are necessary in conjunction to identify causal parameters. The set of matching variables above exhausts the entire set of possible selection criteria for breeders; no unobserved variable can influence mating decisions. Figure 6 shows the degree of overlapping between superstars and alternative sires; since there is a considerable difference in the size of the groups, superstars make up only a tiny fraction of bulls; however, the overlapping is good in the range 0.05 – 0.50, but there exist a few superstars that had few close matches in the sample.

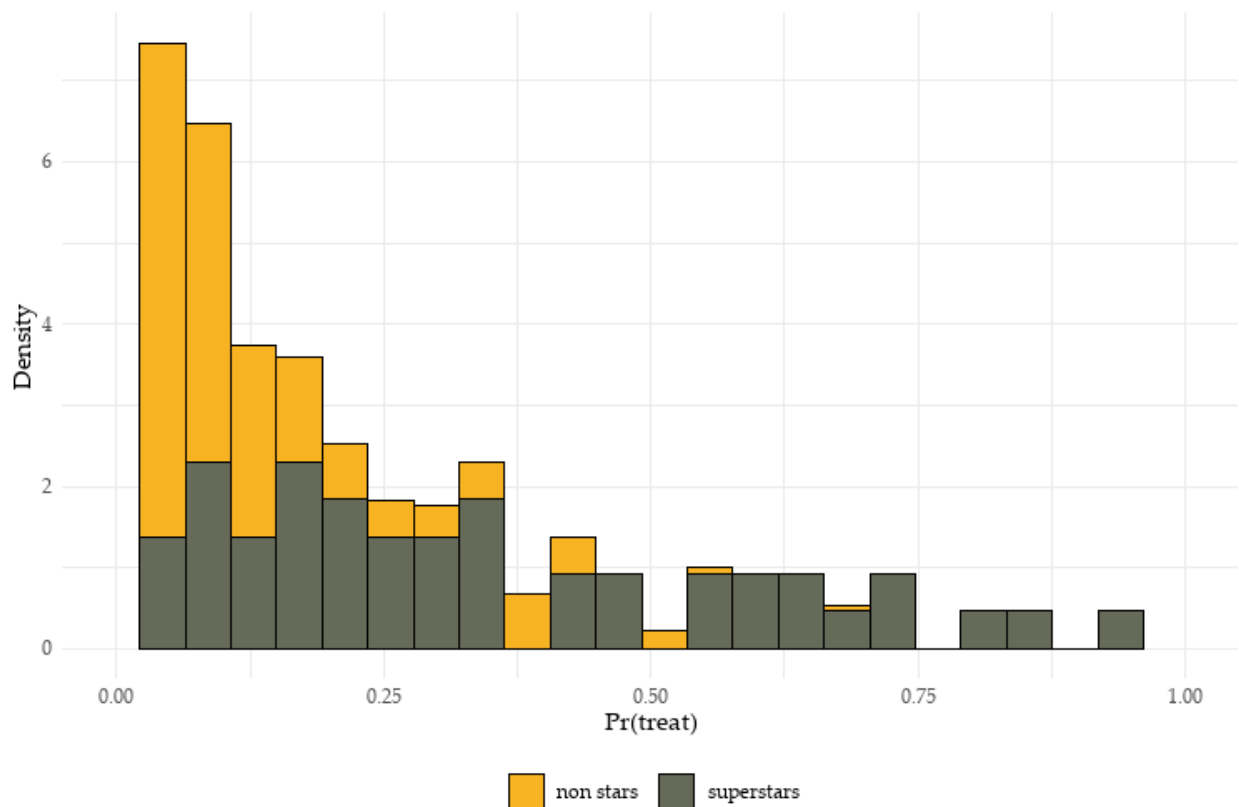


Figure 6: Propensity score by class of bull

A central part of our argument is that treatment and control groups should have similar inbreeding levels before genomic selection. To test that assumption, in Table 1, we calculate the difference in averages of treatment and control groups for inbreeding rates and a few selected traits. There is no statistically significant difference for most traits between treatment and control groups, so our matching method ensures that the difference between both groups is due to the “prestige” of superstar lines because the bulls in the treatment group had a larger offspring than those in the control group. Each year these bulls will have a new set of descendants being born, each with its inbreeding rate and genetic traits; if our hypothesis holds, then the inbreeding rates of the descendants of supersires should be significantly higher than those of non-superstar descendants.¹

¹In Appendix A, we discuss three more assumptions that must hold for our estimator to be valid including parallel trends, no anticipatory effects, and asymptotic convergence.

Table 1: Balance table for outcome and matched covariates

Variable	Control mean (n = 224)	Treatment mean (n = 43)	t.statistic	p.value	df
Inbreeding rate	2.968	2.904	0.426	0.672	58.147
Milk	0.059	0.059	-0.036	0.971	59.956
Fat	-0.122	-0.170	0.346	0.730	76.548
Protein	-0.851	-1.009	0.416	0.679	61.059
Somatic cell score	-0.151	-0.037	-0.675	0.502	66.764
Productive life	-0.619	-0.677	0.173	0.864	59.829
Daughter preg. rate	-1.224	-0.489	-1.336	0.186	70.101
Heifer conc. rate	-11.953	-5.553	-1.807	0.075*	71.841
Cow conc. rate	-0.420	-0.209	-1.278	0.206	65.480
Livability	-0.393	-0.177	-1.816	0.074*	70.905
Type	-0.827	-0.551	-1.721	0.090*	69.510
Gest. length	0.342	-0.055	1.719	0.090*	64.087
Heifer liv.	-0.569	-0.579	0.032	0.975	58.933
Eff. calving	0.101	0.232	-1.549	0.127	58.339
Mastitis	-0.888	-0.764	-0.288	0.774	58.621
Metritis	-0.840	-0.891	0.202	0.840	62.094
Strength	-0.589	-0.581	-0.047	0.963	57.734
Rear legs rear view	-191.265	-24.936	-1.581	0.119	67.713
Rear legs side view	-0.027	-0.013	-0.495	0.623	60.964
Foot leg score	-1.348	-1.123	-0.666	0.508	59.099
Teat rear place	-8.284	-3.255	-1.859	0.067	68.421
Rump angle	-0.360	-0.136	-1.447	0.153	64.334
Thurl width	-0.010	0.050	-0.342	0.734	63.537
Foot angle	-0.767	-0.417	-2.044	0.045**	68.106
Fore udder	-0.743	-0.432	-1.735	0.087*	66.269
Rear udder height	0.142	0.193	-0.254	0.800	60.330
Rear udder width	3.044	2.995	0.690	0.493	52.692
Udder cleft	-0.386	-0.349	-0.235	0.815	77.188
Udder depth	-0.206	-0.267	0.409	0.684	67.108
Teat fron place	-0.343	-0.195	-1.006	0.318	66.497
Teat length	0.197	0.159	0.195	0.846	59.002
Milk fever	-0.312	-0.140	-0.948	0.347	62.811
Stature	-0.255	-0.233	-0.138	0.890	72.664
Dairy form	-0.573	-0.403	-1.442	0.153	73.087
Body depth	-0.401	-0.193	-1.127	0.264	61.777
Sire calv. ease	-0.738	-0.589	-0.989	0.326	73.000
Daughter calv. ease	2.427	2.287	1.188	0.239	63.963

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

We observed the average inbreeding rates of the treatment and matched control groups before and after introducing genomic selection. If we divide the 2005-2017 period into two groups, pre-intervention (2005-2009) and post-intervention (2010-2018), denoted as 1 and 2 and let T_j denotes whether a bull from line j belongs to treatment ($T_j = 1$) or control ($T_j = 0$) groups, then:

$$inb_{jt} = T_j inb_{jt}^{treat} + (1 - T_j) inb_{jt}^{control}$$

Where $j = 1, \dots, J$ are the lines, each comprised of n_j bulls such that $i = 1, \dots, n_j$. To measure the effects of genomic selection on inbreeding, we estimate the differences-in-differences model:

$$inb_{ikt} = treat_i + treat_i * \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(yob = t) + \alpha_t + \alpha_k + \varepsilon_{ijt} \quad (\text{Model 2})$$

$$inb_{ikt} = treat_i + treat_i * \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(yob = t) + \alpha_t + \alpha_k + \mathbf{X}_{it} \gamma + \varepsilon_{ijt} \quad (\text{Model 3})$$

$$inb_{ikt} = treat_i + treat_i * \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(yob = t) + \alpha_t + \alpha_k + \mathbf{X}_{it} \gamma + (\mathbf{X}_{it} * post_t) \rho + \varepsilon_{ijt} \quad (\text{Model 4})$$

Model (2) is a specification that does not contain any covariates (PTAs), and it decomposes the ATT on inbreeding rates by year of birth cohort. Model (3) incorporates the bull's genetic traits ($\mathbf{X}_{it}$) and firm-level fixed effects, and Model (4) includes interactions of genetic traits and the post-treatment dummy. We include the last specifications to account for trait-level trends. We also know that the treatment affects both inbreeding rates and PTAs, but we do not have a structural model for their impact (Lewbel, 2019), so we include them in a non-linear fashion as both levels and interactions to reduce the uncertainty about the functional form of our controls.

5 Results

Figure 7 shows treatment and control groups' average inbreeding rates and confidence intervals across years. Genomic selection did not increase inbreeding rates immediately; both curves began to diverge substantially after 2011. Inbreeding rates, in turn, started growing two years later, and both curves diverged significantly from 2012 onwards.

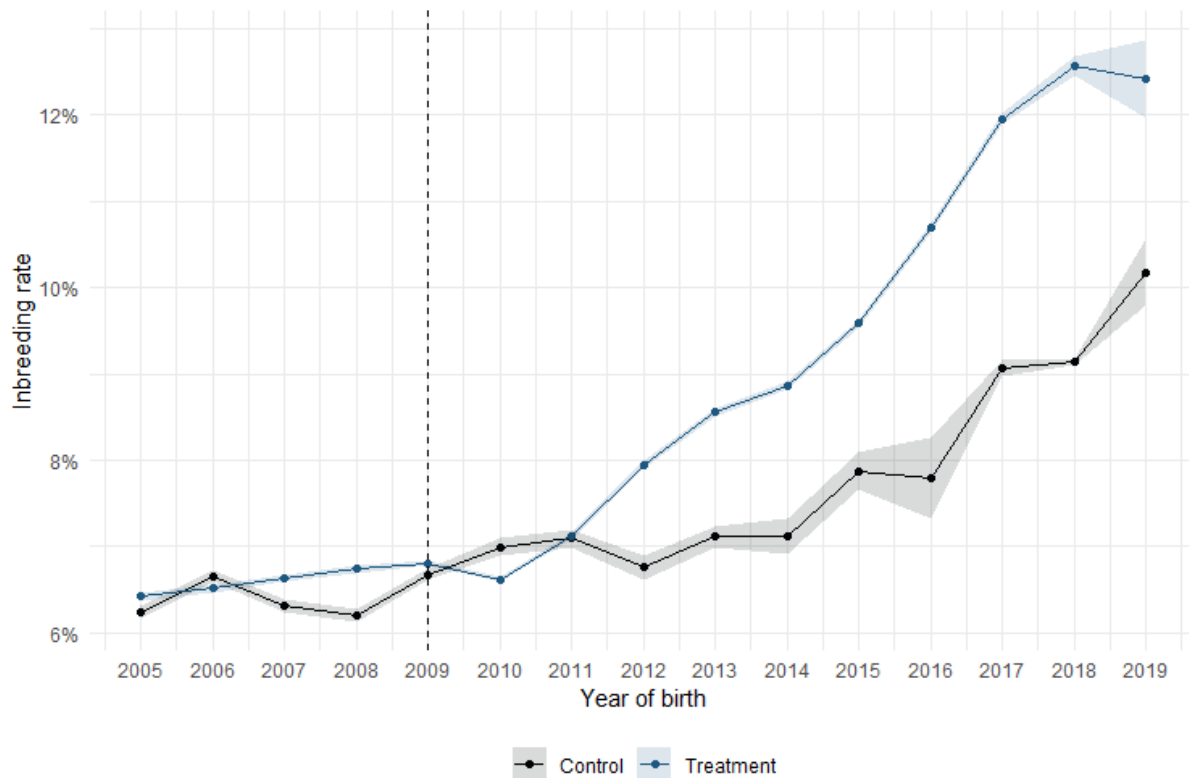


Figure 7: Average inbreeding rates in treatment and control groups

This plot closely matches the evolution of the population average inbreeding rates plotted in Figure 2. This behavior is consistent with our hypothesis that breeders prefer well-known over new lines. Our findings coincide with results reported by Guinan et al. (2023), who also look at the time trends of productivity and reproduction traits in US dairy cattle. The authors also document a decrease in the generation interval and an increase in inbreeding rates of Holstein and Jersey cattle but do not consider the latter an issue because the expected inbreeding rates of future offspring adjust genetic trait values.

Table 2: Difference-in-Differences coefficient estimates

	Inbreeding Rate			
	No covariates	Traits	Traits and interactions	
	(1)	(2)	(3)	(4)
Treat $\times$ Post = 1	1.453*** (0.4250)			
Treat $\times$ YoB = 2005		0.0944 (0.2975)	0.0489 (0.2989)	0.0252 (0.2829)
Treat $\times$ YoB = 2006		-0.2183 (0.3067)	-0.1182 (0.2933)	-0.2789 (0.2534)
Treat $\times$ YoB = 2007		0.2328 (0.3038)	0.2826 (0.2676)	0.2848 (0.2534)
Treat $\times$ YoB = 2008		0.4035 (0.3148)	0.3246 (0.2798)	0.3294 (0.2425)
Treat $\times$ YoB = 2010		-0.4589 (0.2827)	-0.4721* (0.2826)	-0.5583** (0.2667)
Treat $\times$ YoB = 2011		-0.0164 (0.3599)	-0.0455 (0.2596)	-0.1671 (0.2511)
Treat $\times$ YoB = 2012		1.224** (0.4906)	1.252** (0.5633)	1.151* (0.5981)
Treat $\times$ YoB = 2013		1.433*** (0.4124)	1.083** (0.4631)	0.9980** (0.4725)
Treat $\times$ YoB = 2014		1.780*** (0.3017)	1.600*** (0.3514)	1.501*** (0.3068)
Treat $\times$ YoB = 2015		1.637*** (0.3453)	1.363*** (0.4159)	1.255*** (0.4256)
Treat $\times$ YoB = 2016		2.414*** (0.2424)	1.766*** (0.2697)	1.710*** (0.2564)
Treat $\times$ YoB = 2017		2.849*** (0.4456)	2.770*** (0.4429)	2.734*** (0.4507)
Treat $\times$ YoB = 2018		3.417*** (0.4530)	3.576*** (0.4566)	3.515*** (0.4847)
Treat $\times$ YoB = 2019		2.005** (0.8611)	2.071** (0.8827)	2.022** (0.8785)
p-value for nonzero pre-effect		0.416	0.498	0.107
Observations	15,491	15,491	15,491	15,491
R ²	0.16988	0.47369	0.49942	0.50471
Within R ²	0.00702	0.01235	0.04150	0.05163
Treat fixed effects	✓	✓	✓	✓
Post fixed effects	✓			
Year of birth fixed effects		✓	✓	✓
Company fixed effects			✓	✓

Table 2 shows the results of three models. Model (1) only includes the interaction between treatment and an indicator for the post period, making it a 2x2 difference-in-difference model. Model (2) splits the treatment effect by year of birth averages, Model (3) includes the complete set of genetic traits and firm-level fixed effects, and Model (4) is the model with traits and a post-treatment dummy. In all specifications, standard errors are clustered at the line level to better deal with the inconsistency generated by the autocorrelation of inbreeding rates between different generations (Bertrand et al., 2004).

Model (1) is the simple differences-in-differences estimation where $post_t$ is a dummy equal to 1 if the bull was born after 2009. The results show that bulls descending from supersires increased their inbreeding rates by 1.46 percent more than those who descend from less popular bulls, on average. Since the impact of genomic testing on inbreeding rates only became noticeable after 2011, the coefficient underestimates the average treatment effect on the treated (ATT) of genomic selection. Model 2 shows that the treatment-year-of-birth interactions are positive and significant from 2012 onwards, with the 2018 effect being the highest.

Models (3) and (4) in Table 2 show that there is a positive and significant incremental variation for the treated versus control groups in the years 2012-2018 even with the inclusion of controls. In all years after 2010, the coefficients are positive. They both show a positive and increasing inbreeding rate in the treatment group comprised of all descendants from prestige sires relative to the control group of descendants of comparable non-superstar sires.²

Table 2 also includes the results for the F-test for joint significance of all pre-treatment differences in trends for the interaction of treatment and year of birth. The p-values show no significance at any specific level in any of the specifications.³

²The last period in the sample is 2018 instead of 2020 because we cannot make inferences after 2018 due to a reduction in the number of animals in the control sample; the number of comparable descendants is too small for us to consider them a valid comparison group. This drastic reduction is a consequence of lower fertility rates in the control group; since those sires fathered fewer sons, their sons will beget even fewer grandsons, and after a few years, this group collapses.

³This is not a test for parallel trends since the assumption relies on unobservable counterfactuals. Rather,

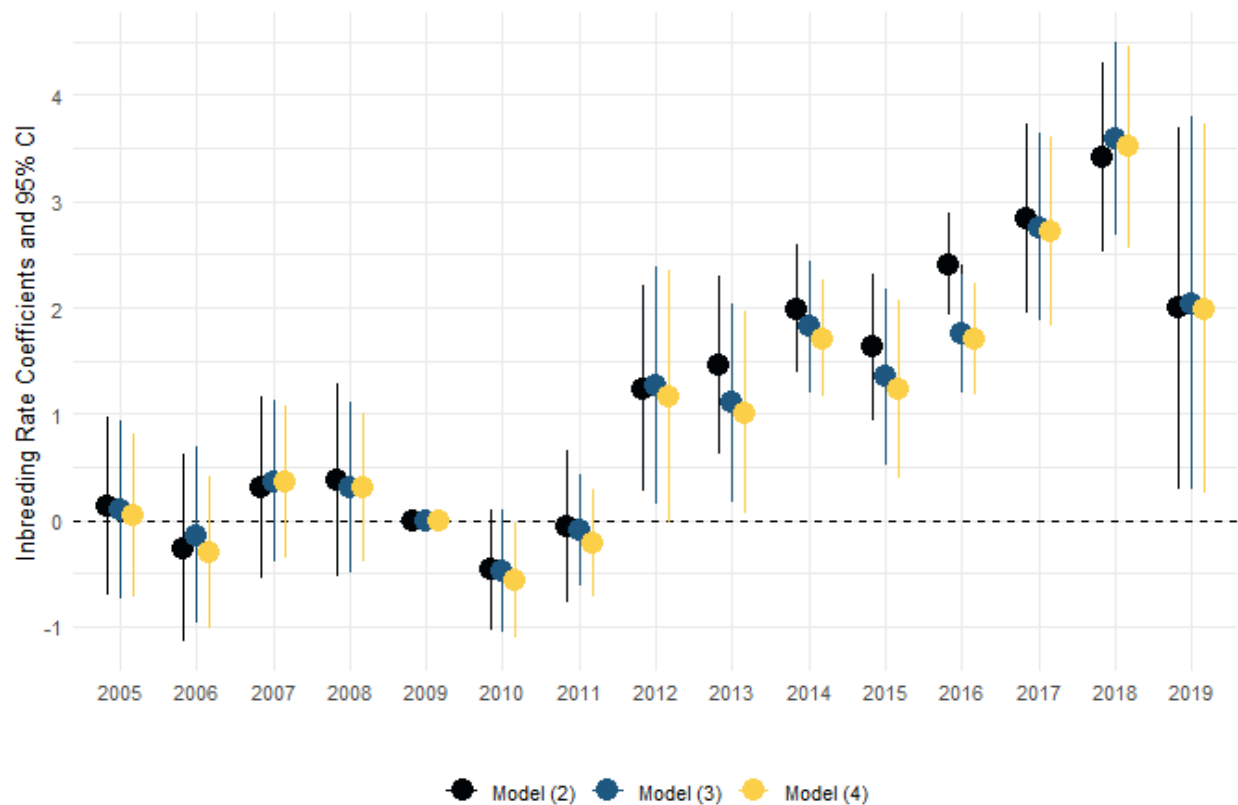


Figure 8: Coefficient plots for different specifications

Figure 8 shows the coefficients from the interaction of year fixed effects and treatment status to assess the impact of inbreeding per year. Every year after 2012, the inbreeding rate of the treatment group is between 1 and 2.8 percent points higher than the control group. The coefficient values decrease slightly after controlling for their genetic traits but remain statistically significant. The figure also shows that all pre-treatment coefficients are not significant. Figure 8 uses the *sup-t* uniform confidence intervals for the pre-treatment periods of Montiel-Olea and Plagborg-Møller (2019) as suggested by Freyaldenhoven et al. (2019) instead of the standard pointwise confidence. These confidence intervals contain the actual path of the coefficients 95% of the time and can better help understand which types of pre-trends are consistent with the data.

it is usually considered evidence in favor of the assumption of parallel trends. However, this is not necessarily the case since this test has low power against a linear or quadratic pre-trend alternative (Roth, 2022). In our case, we can dismiss this concern; since our matching method ensures the fulfillment of the condition for the founders, the entire effect must be due to their sons and grandsons.

6 Cost-Benefit Analysis

Using published estimates of the costs of inbreeding, we can attach a monetary value to increased inbreeding rates. Smith et al. (1998) shows that every additional inbreeding percent is associated with 24.45 USD less income by milk sales. The total cost depends on what we consider the average inbreeding. We can use the control group rates as a counterfactual and calculate the cost from increased inbreeding of the treatment group, multiplied by the total stock of Holstein cattle. In that instance, we estimate a total cost from increased inbreeding of 3.7 billion dollars for the period 2011-2019. Alternatively, by industry standards, the maximum recommended inbreeding rate could be 6.25%⁴. Calculating the cost from inbreeding levels greater than such value, we estimate the total cost for the same period as 7.2 billion dollars for the entire dairy industry.

While genomic testing increased inbreeding, it also increased milk production and thus profits for the dairy industry. The benefits of introducing genomic selection to the dairy industry can be calculated using an index called **Net Merit**, which measures the net profit over the lifetime of the animal's average daughter, expressed in U.S. dollars. Net merit is calculated by multiplying the values of genetic traits by a set of dollar values intended to capture the "market value" of all traits.⁵

⁴<https://www.agproud.com/articles/36810-the-impact-of-genomics-on-rapid-increase-of-inbreeding-of-holsteins>

⁵Table 3 shows what are the magnitudes of such weights (Van Raden et al., 2021). Note that the calculation involves the usage of subindexes for health and reproduction traits, as well as an estimation of residual feed intake per animal.

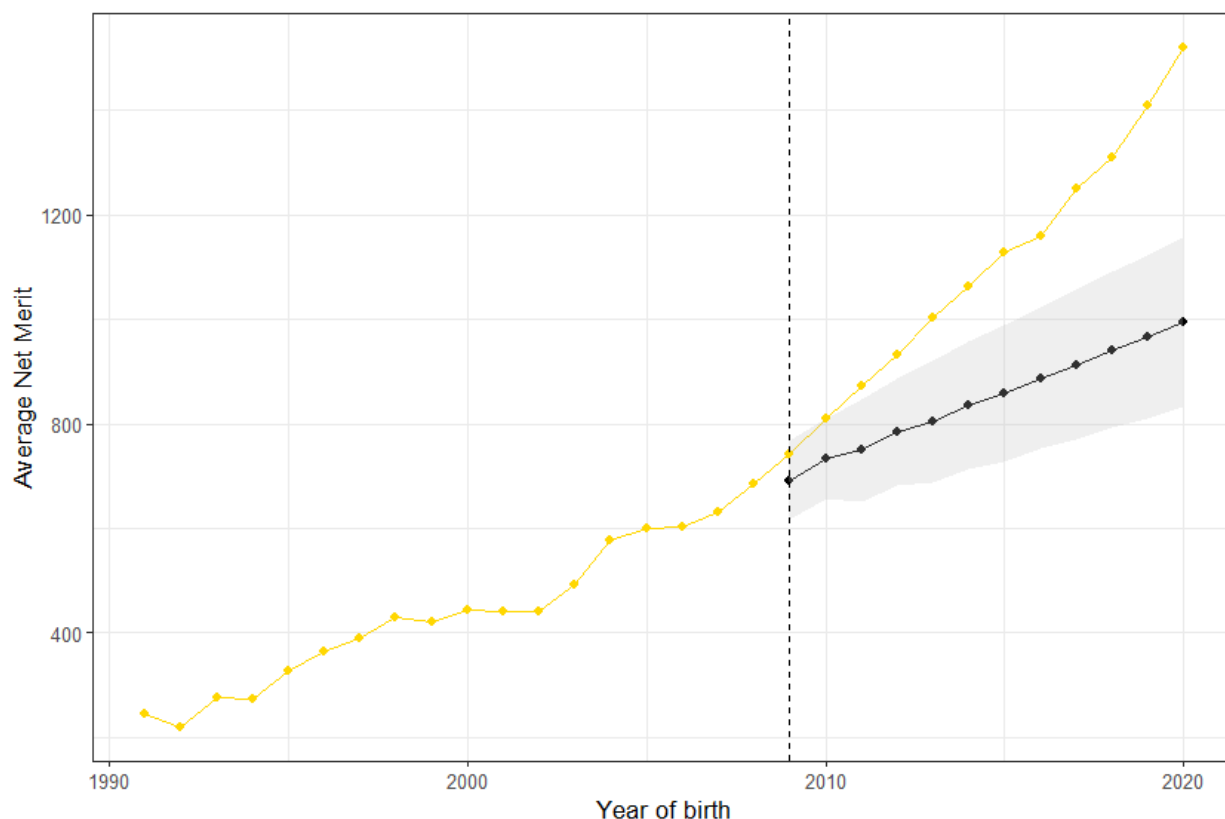


Figure 9: Observed and Predicted Net Merit Values

To approximate the impact of genomic testing on Net Merit, we look at its pre- and post-2009 trends. We use the data from 1985 to 2009 to fit an ARIMA model to predict its values for the out-of-sample period of 2010-2020⁶. Figure 9 shows average Net Merit values per year (yellow line) and its predicted values from 2009 (black line). Starting in 2011, the performance of the Net Merit index vastly outperforms even the 95% prediction interval from this model, and it amounts to an average of 275 USD per animal over the 11 years compared to the predicted (pre-introduction) values.

These numbers can be easily converted into an estimation of total benefits from genomic selection by multiplying the gain in Net Merit by the estimated number of cattle in the United States and dividing this number by one-half (because the gains come from the male side only.) This amounts to 14 billion USD for 2009-2020, or 1.2 billion dollars annually

⁶We use the automatic order selection method described in Hyndman and Khandakar (2008), which estimates a series of models and then selects the best model on the basis their Akaike Information Criterion

on average. In terms of net benefits, this implies that the benefits of genomic testing net the costs of inbreeding equals 6.9-10.42 billion US dollars.

7 Discussion

This article investigates the consequences of genomic selection on the US dairy industry. On the one hand, this new technology significantly improved productivity. On the other hand, we find evidence that it increased the average inbreeding rate of all bulls in the market. As genomic testing increased the rate of genetic improvement, it appeared to have exerted competitive pressure on animal breeders to select from popular lines where the gains are high. By not crossing enough from other lines, the inbreeding rate increases and runs the risk of introducing future health problems in dairy cattle in the US and around the world. As the average inbreeding rate increases, every new generation of animals experiences an increased risk of experiencing inbreeding depression and a higher chance of genetic diseases hidden in recessive traits.

Because dairy cattle breeders are not incentivized to cross-breed in the current market, this suggests a role for the government or an industry group to assist in solving the coordination problem. Currently, the Holstein Association of the United States adjusts the values of genetic traits according to inbreeding levels to penalize breeders that select bulls with high inbreeding rates to account for the higher reproduction costs of such animals. Yet, Cole (2023) points out that many of the most inbred bulls have such high levels of traits that the downgrading will not temper the demand for them enough. Another option would be to give dairy farmers more information about the dangers of inbreeding to reduce demand for the most inbred lines (Cole, 2023).

Similar coordination problems have been confronted in the application of pesticide and use of Bt-resistant crops by farmers. In the case of Bt-resistant crops, the US EPA and many countries decided on a command and control strategy which mandates farmers

to plant some area to non-Bt-resistant crops (Huang et al., 2011). This amounts to a tax, in the form of a yield penalty, on farmers to force them to internalize the costs of their actions. Whether such a strategy would work for dairy cattle breeding would depend on some key factors. Namely, how logistically might a government or organization impose a similar penalty on dairy cattle breeders to steer them away from inbreeding?

More research is needed in this area in agricultural economics, as the coordination problems of dairy cattle breeding have not yet been examined by the field. Yet, the field has exactly the theoretical frameworks for examining these kinds of problems in the dairy industry. Moreover, studying the increase in inbreeding is crucial for ensuring a safe and global resilient dairy sector in the future and ensuring that US dairy cattle breeders continue to sustainably raise the productivity of dairy cattle in the future.

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A Additional Estimator Assumptions

Our causal estimand of interest is the Average Treatment on the Treated (ATT) in the post-intervention period:

$$ATT_{post} = E \left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1 \right]$$

This quantity measures the average causal effect on the treated lines in the post-treatment period (Roth et al., 2023). To correctly identify causal effects, a series of assumptions must hold (Sun and Abraham, 2021):

Assumption 3 (Parallel trends):

$$E \left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 1 \right] = E \left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 0 \right]$$

We assume that the average inbreeding rates for both groups would have evolved in parallel without introducing genomic selection. This assumption is not testable because we cannot observe untreated inbreeding rates after 2009, but it constrains our model specification to a Two-Way Fixed Effects specification we call Model (1):

$$inb_{ikt} = treat_i + (treat_i * post_t)\beta + post_t + \alpha_k + \varepsilon_{ijt} \quad (\text{Model 1})$$

Where inb_{ikt} is the inbreeding rate of bull i born from a line that belongs to either the treatment or control group born in year t , supplied by firm k , β_t is a year of birth fixed effect, and α_j is a firm fixed effect.

Assumption 4 (No anticipatory effects): $inb_{pre}^{control} = inb_{pre}^{treat}$

This assumption ensures that both groups were comparable before the treatment; inbreeding rates of superstar lines did not increase before the introduction of genomic selection; we use matching methods to ensure that this condition holds.

Under Assumptions 3 and 4, we have that:

$$ATT_{post} = E \left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1 \right] - E \left[inb_{pre}^{treat} - inb_{pre}^{control} \middle| T_j = 0 \right] \quad (2)$$

The ATT is identified as the difference-in-differences of observed outcomes in the pre and post-intervention periods. All elements in this equation are observed and, therefore, estimable. Suppose superstars had superior traits to our matched control group. In that case, this condition is violated, and we would not be able to parse out the causal effect of inbreeding from the effect of the bull's genetic superiority. Assumptions 3 and 4 ensure that the OLS estimates $\hat{\beta}$ from Model (2) are consistent and asymptotically valid (Roth et al., 2023).

Assumption 5: Let $W_j = (inb_{j,pre}, inb_{j,post}, T_j)'$ be a vector of outcomes and treatment status for unit i . Then, we observe a sample of N independent and identically distributed random draws $W_i \sim F_W(w)$ for a distribution F satisfying parallel trends.

Under Assumptions 3, 4, and 5: $\sqrt{N}(\hat{\beta} - ATT_{post}) \xrightarrow{D} N(0, \sigma^2)$, then we can consistently estimate the variance σ^2 clustering at the line level (Bertrand et al., 2004) asymptotically ($N \rightarrow \infty$ and T fixed).

We can also decompose the variations in treatment time since every year, a certain number of bulls will be born from lines in both groups, each with its inbreeding rate that can be averaged across treatment and control groups. Models 2-4 show the complete specification with leads and lags of the treated group. These models decompose the estimated ATT relative to the treatment average by birth year. Sun and Abraham (2021) call this specification "Dynamic specification." However, there is no staggered treatment since genomic selection affects all lines in the same year, but inbreeding rates evolve independently after the treatment.

This assumption states that the treatment status of an individual from any given line is not affected by the treatment status of others. In our setting, the treatment (genomic selection) affects all individuals, so there is no selection into treatment.

B Robustness Checks

B.1 Total number of descendants

In our analysis, we defined a “supersire” as a bull whose total number of descendants was among the 95th percentile of their distribution by 2021. The cutoff point also influences this calculation since genomic testing leads to an expansion of the supply of bulls, and the size of any line is endogenous to genomic selection. It can be argued that selecting bulls on the ex-post number of sons is tautological to our argument.

Figure 10 plots the coefficients for the different differences-in-differences specifications, but this time, the number of descendants is calculated only up to 2010, so we assume that genomic selection is just being introduced, and we are trying to guess which lines will be more successful. Coefficients for 2014 and 2015 are now substantially reduced and become insignificant, but the overall result remains unchanged; we still have that the treatment group has notoriously higher inbreeding rates than the control group, as seen in the 2×2 model, where the coefficient is still positive and significant for the entire period, as seen on 4.

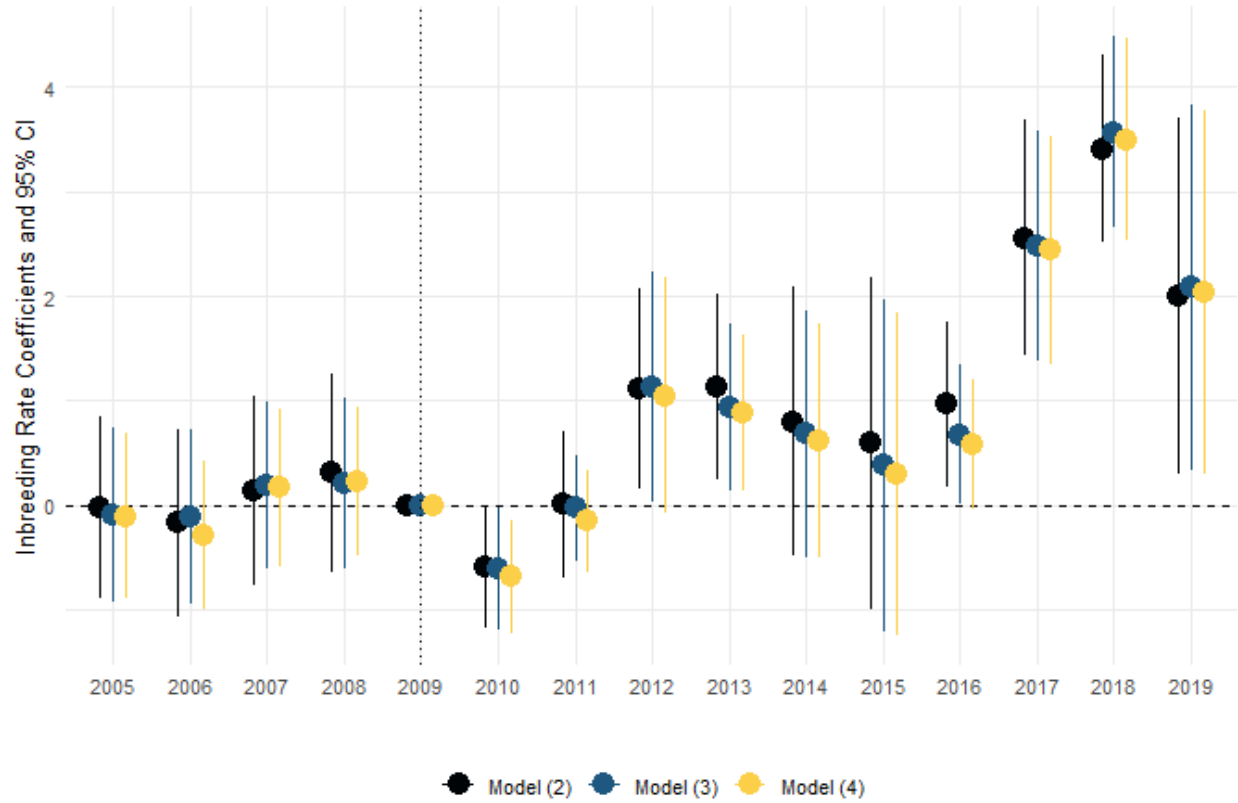


Figure 10: Coefficient plots for an alternative definition of supersire

B.2 Quarter of birth effects

Since we have the exact date of birth of every animal in the sample, we can group them by a quarter of birth as well; Figure 11 shows the results from Models 2-4 but replacing year for a quarter of birth interactions with the treatment. The signs and magnitudes of coefficients do not differ substantially from the original formulation. The increase in coefficient values is monotonic and statistically significant since 2012, but having a smaller sample size per quarter makes confidence intervals wider.

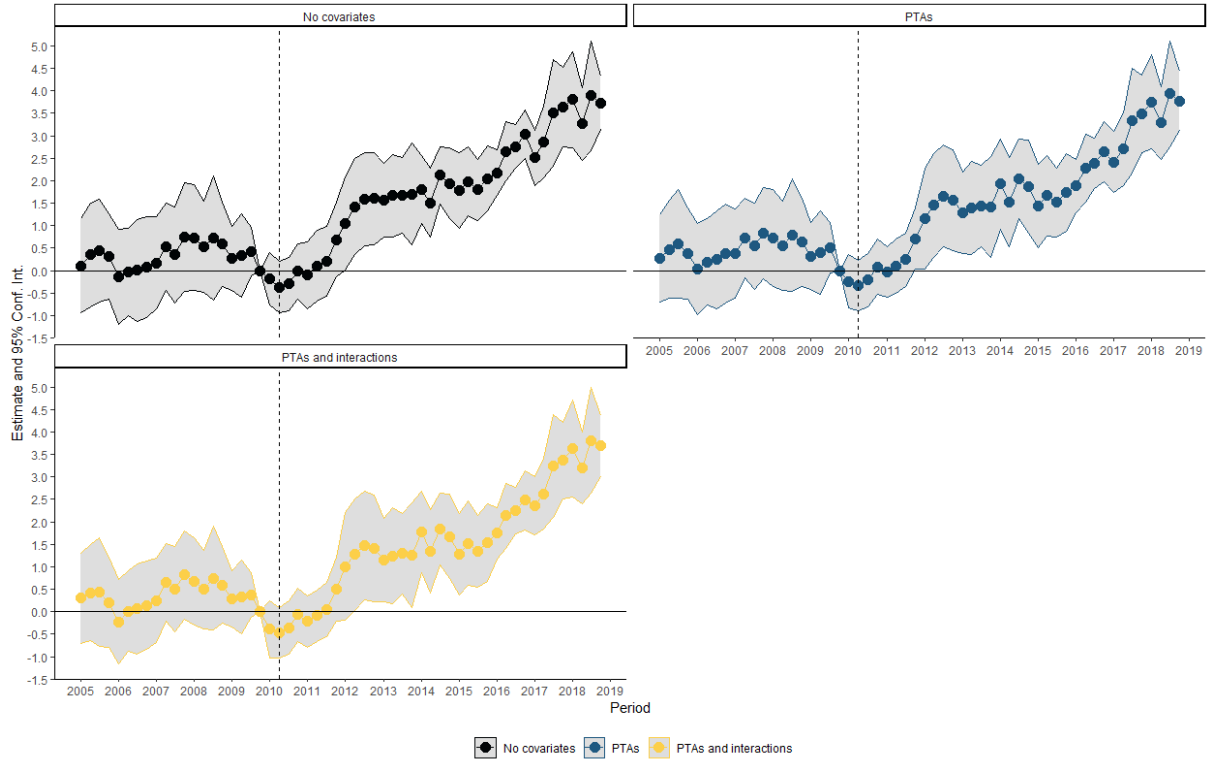


Figure 11: Coefficient plots for quarter of birth effects

This article’s results depend heavily on a somewhat arbitrary definition of a “superstar” bull in terms of the size of his line. What would happen if we changed that from the 95th to another quantile of the line size distribution? Figure 12 shows the coefficient plots for different definitions of “superstar”.

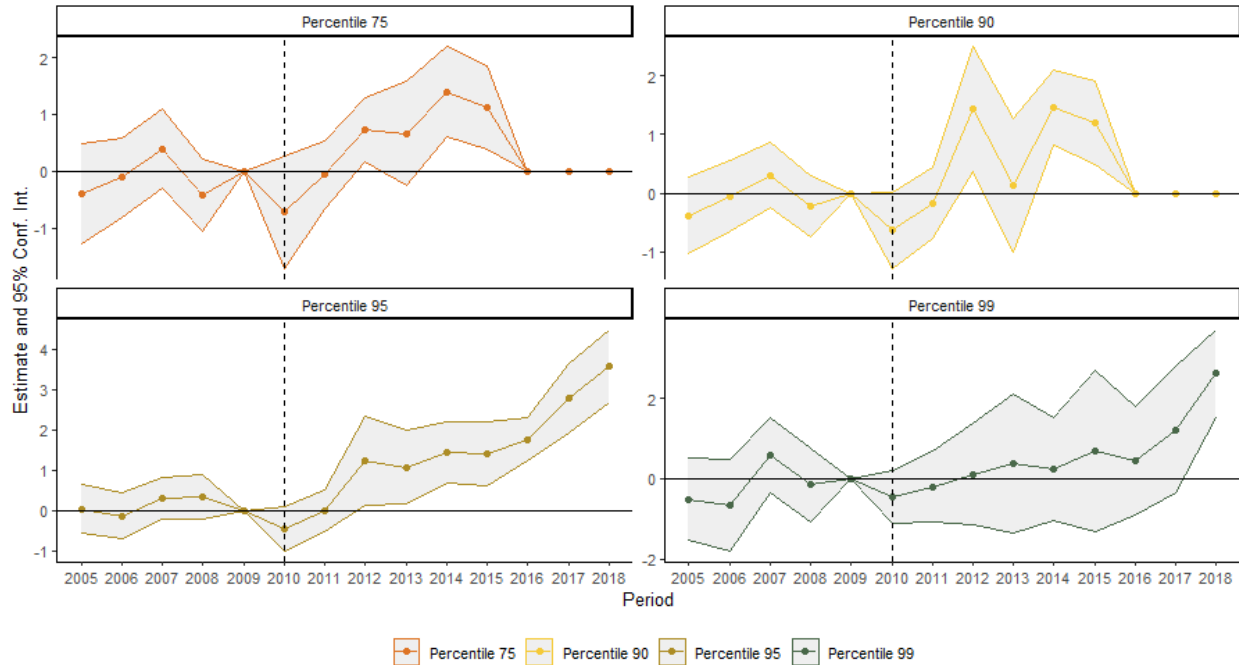


Figure 12: Robustness test: changes in the definition of “superstar”

If we change the cutoff point from the 95th to the 99th percentile, the sign, and magnitude of our coefficients do not change much. Still, the confidence intervals widen significantly because our treatment group comprises such specific animals that the matching algorithm cannot find suitable matches. As a consequence, the significance of the difference-in-differences coefficients decreases significantly. Going in the other direction, we can find marginally better matches for the superstar group. However, the control group ended abruptly in 2015 because most of the original group’s control lines are now included in the treatment, making the control group short-lived. We conclude that our results depend critically on a set of lines that allow for a good comparison when included in the control group; otherwise, the analysis yields insignificant results.

B.3 Date of birth of “superstars”

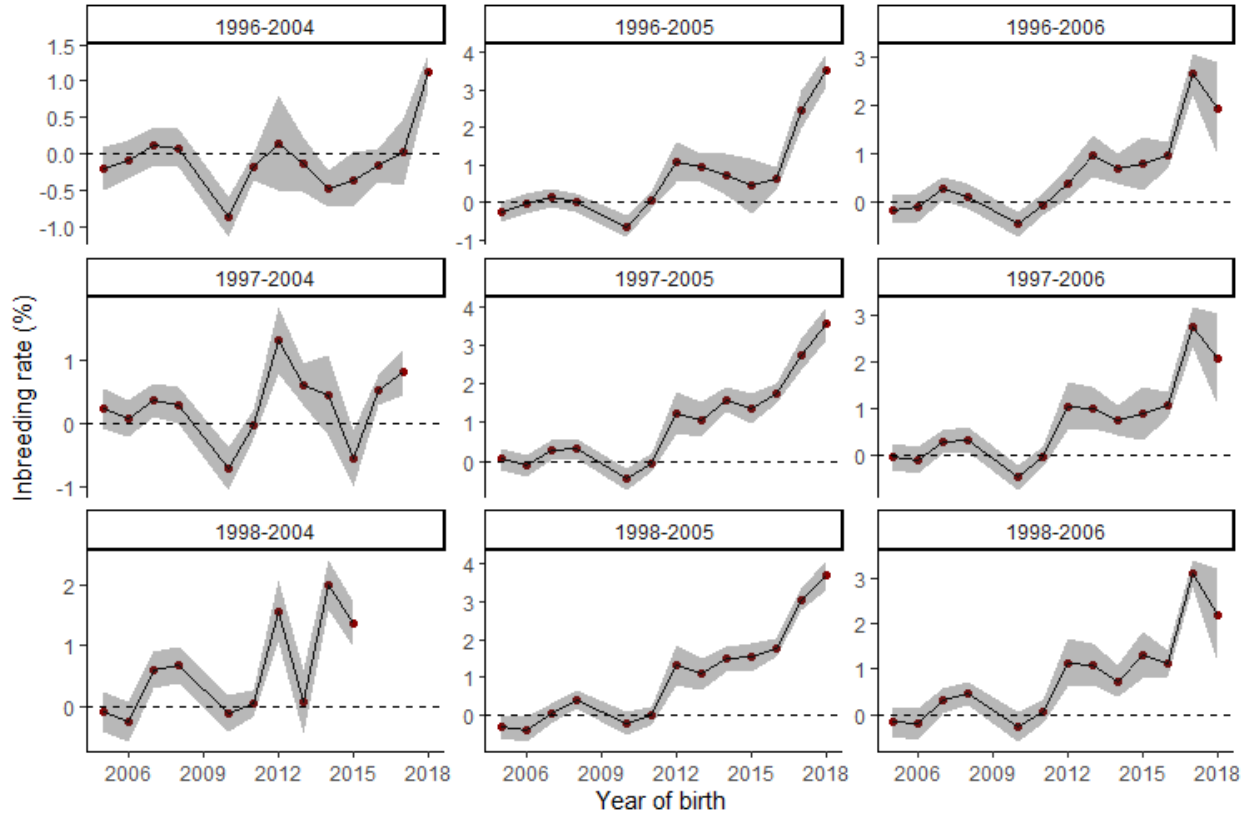


Figure 13: Coefficients from model (2) per different definitions of founders’ date of birth

Finally, Figure 13 shows coefficient plots by different definitions of the founders’ date of birth. We defined founders as all bulls born between 1997 and 2005, but such a definition is also arbitrary. If we adopt different cutoffs for the period where the founders were born from 1996/1998 to 2004/2006, we have $3 \times 3 = 9$ distinct combinations for starting and end date. The coefficient plot shows that the bulk of the effect can be attributed to sires born from 1996 to 2006; if we extend further the selection period of sires, we will have more endogeneity issues since, by this date, it was known that the bovine genotype was being sequenced, and thus, that there would be a significant effect on the market.

C Additional Plots

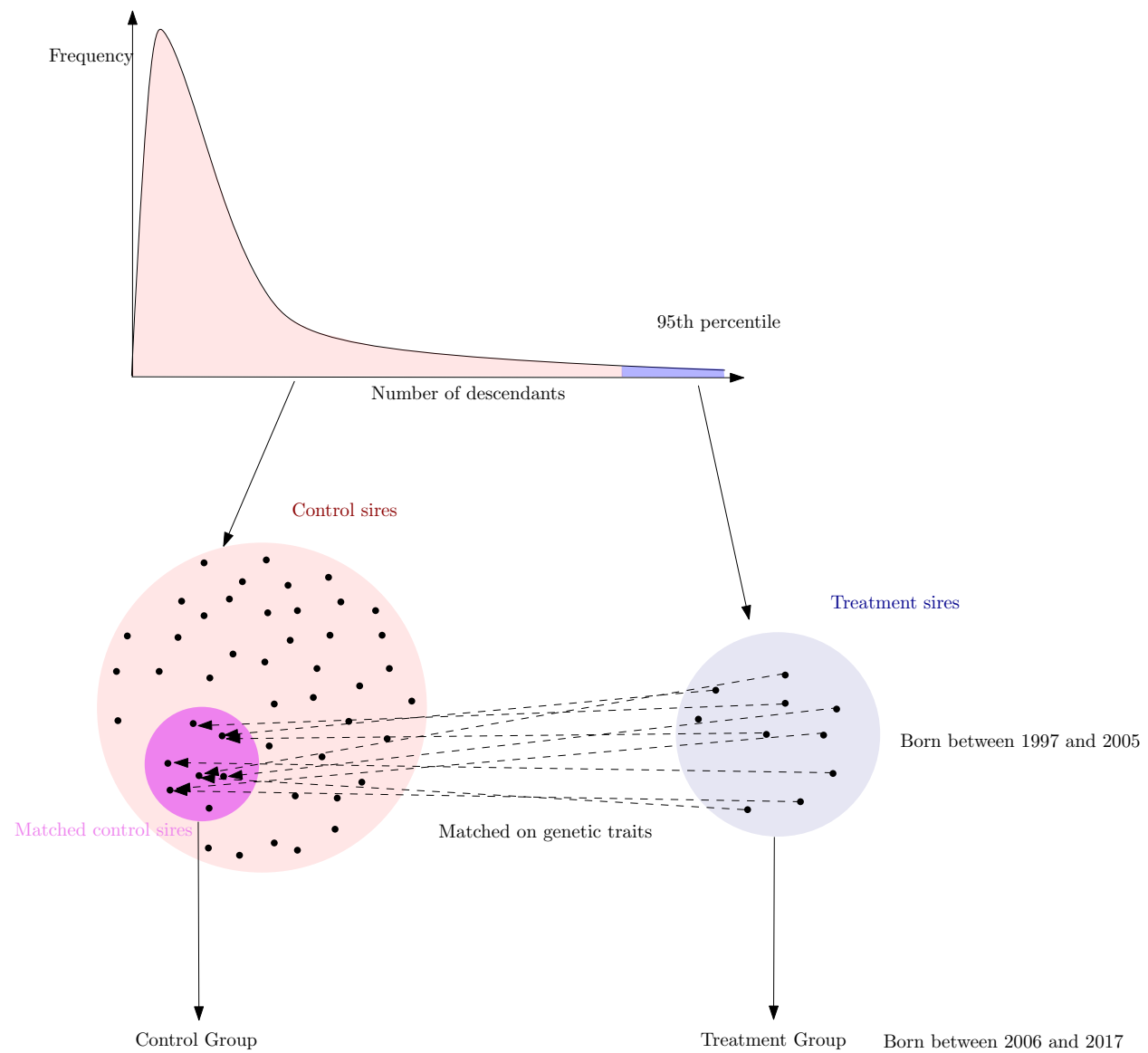


Figure 14: Sample selection mechanism

Trait	Units	Value (\$/PTA unit)
Milk	Pounds	0.002
Fat	Pounds	4.18
Protein	Pounds	4.67
Productive Life	Months	34
Somatic Cell Score	Log	-74
Body Weight Composite	Composite	-45
Udder Composite	Composite	19
Foot/Leg Composite	Composite	3
Daughter Pregnancy Rate	Percent	11
Calving Trait Subindex	USD	1
Heifer Conception Rate	Percent	1.1
Cow Conception Rate	Percent	2.2
Livability	Percent	9.8
Health Trait Subindex	USD	1.0
Residual Feed Intake	Pounds	46.2
Effective Calving	Days	2.1
heifer Livability	percent	5.0

Table 3: Economic values for **Net Merit**

Table 4: Difference-in-Differences estimates from alternative scenario

	No covariates		inbreeding	
	(1)	(2)	Traits (3)	Traits and interactions (4)
treat $\times$ post = 1	1.217*** (0.4671)			
treat $\times$ yob = 2005		-0.0189 (0.2991)	-0.0863 (0.2966)	-0.1019 (0.2842)
treat $\times$ yob = 2006		-0.1702 (0.3088)	-0.1060 (0.2951)	-0.2813 (0.2520)
treat $\times$ yob = 2007		0.1376 (0.3131)	0.1949 (0.2826)	0.1698 (0.2699)
treat $\times$ yob = 2008		0.3104 (0.3286)	0.2154 (0.2896)	0.2295 (0.2508)
treat $\times$ yob = 2010		-0.5930** (0.2916)	-0.6049** (0.2935)	-0.6837** (0.2705)
treat $\times$ yob = 2011		0.0069 (0.3567)	-0.0263 (0.2577)	-0.1519 (0.2509)
treat $\times$ yob = 2012		1.115** (0.4877)	1.134** (0.5545)	1.052* (0.5685)
treat $\times$ yob = 2013		1.131** (0.4479)	0.9319** (0.4026)	0.8830** (0.3758)
treat $\times$ yob = 2014		0.8006 (0.6513)	0.6818 (0.6000)	0.6248 (0.5656)
treat $\times$ yob = 2015		0.5968 (0.8029)	0.3803 (0.8051)	0.3005 (0.7810)
treat $\times$ yob = 2016		0.9684** (0.3983)	0.6753** (0.3389)	0.5804* (0.3175)
treat $\times$ yob = 2017		2.558*** (0.5683)	2.484*** (0.5576)	2.437*** (0.5518)
treat $\times$ yob = 2018		3.407*** (0.4524)	3.567*** (0.4620)	3.493*** (0.4911)
treat $\times$ yob = 2019		1.995** (0.8613)	2.083** (0.8848)	2.035** (0.8804)
p-value for nonzero pre-effect		0.772	0.799	0.414
Observations	15,352	15,352	15,352	15,352
R ²	0.16918	0.47434	0.50014	0.50506
Within R ²	0.00575	0.01144	0.04123	0.05067
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓